

Diversity of Pectoral Fin Structure and Function in Fishes With Labriform Propulsion

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ABSTRACT Aquatic propulsion generated by the pectoral fins occurs in many groups of perciform fishes, including numerous coral reef families. This study presents a detailed survey of pectoral fin musculoskeletal structure in fishes that use labriform propulsion as the primary mode of swimming over a wide range of speeds. Pectoral fin morphological diversity was surveyed in 12 species that are primarily pectoral swimmers, including members of all labroid families (Labridae, Scaridae, Cichlidae, Pomacentridae, and Embiotocidae) and five additional coral reef fish families. The anatomy of the pectoral fin musculature is described, including muscle origins, insertions, tendons, and muscle masses. Skeletal structures are also described, including fin shape, fin ray morphology, and the structure of the radials and pectoral girdle. Three novel muscle subdivisions, including subdivisions of the abductor superficialis, abductor profundus, and adductor medialis were discovered and are described here. Specific functional roles in fin control are proposed for each of the novel muscle subdivisions. Pectoral muscle masses show broad variation among species, particularly in the adductor profundus, abductor profundus, arrector dorsalis, and abductor superficialis. A previously undescribed system of intraradial ligaments was also discovered in all taxa studied. The morphology of these ligaments and functional ramifications of variation in this connective tissue system are described. Musculoskeletal patterns are interpreted in light of recent analyses of fin behavior and motor control during labriform swimming. Labriform propulsion has apparently evolved independently multiple times in coral reef fishes, providing an excellent system in which to study the evolution of pectoral fin propulsion. *J. Morphol.* 000:000–000, 2005. © 2004 Wiley-Liss, Inc.

KEY WORDS: pectoral fins; Labroidei; functional morphology; musculoskeletal system; reef fishes; locomotion

Roughly 15–20% of living fishes use their pectoral fins as their primary mode of locomotion (Westneat, 1996), including cichlids, damselfishes, parrotfishes, wrasses, surperches, and many of the angelfishes, butterflyfishes, goatfishes, surgeonfishes and other coral reef families. In addition to their renowned ecological diversity (Randall et al., 1990), fishes with labriform propulsion employ a diversity of flapping and rowing fin movements that vary on a continuum (Walker and Westneat, 2002) from rowing the fins back and forth to flapping the fins up and down. Recent observations have suggested that there may also be considerable variety among reef fishes in the

functional morphology of this mode of locomotion (Westneat, 1996; Walker and Westneat, 2002).

Studies of pectoral fin locomotion have often focused on fin kinematics (Webb, 1973; Blake, 1979; Geerlink, 1983, 1989; Gibb et al., 1994; Drucker and Jensen, 1996; Walker and Westneat, 1997) to describe frequency, amplitude, and mechanisms of thrust using this locomotor mode. Several authors have examined pectoral muscle activation patterns to develop more detailed functional analyses of fin mechanics (Drucker and Jensen, 1997; Westneat and Walker, 1997). Fin motion has also been combined with theoretical, experimental, and computational approaches to analyze pectoral fin hydrodynamics (Blake, 1981, 1983; Lauder and Jayne, 1996; Drucker and Lauder, 2000; Ramamurti et al., 2002). Finally, several studies have examined pectoral fin structure in the context of fin kinematics (Geerlink, 1983, 1989; Westneat, 1996), and more recent workers have examined fin shape (Bellwood and Wainwright, 2001; Fulton et al., 2001; Wainwright et al., 2002; Walker and Westneat, 2002). Many important features of pectoral fin behavior and biomechanics have been outlined, but little information is available on the morphological diversity of pectoral fins among fishes.

Research on behavior, structure, and hydrodynamics of the pectoral fins generates predictions regarding correlations between fin structure and function (Westneat, 1996; Walker and Westneat, 1997, 2002). Pectoral fin mechanisms function along a continuum, from a fore-aft rowing stroke used for high transient thrust and maneuverability, to a dorsoventral flapping stroke that is characteristic of high speed, endurance, and efficiency. Although pec-

Contract grant sponsors: Field Museum of Natural History (scholarship program), National Science Foundation; Contract grant number: IBN 0235307; Contract grant sponsor: Office of Naval Research; Contract grant number: N00014-99-1-0184.

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Published online in
Wiley InterScience (www.interscience.wiley.com)
DOI: 10.1002/jmor.10173

TABLE 1. Reef taxa surveyed indicating muscle insertion patterns on medial fin rays

PCA ID	Genus	Species	ADP	ADR	Rays	NB rays
1	<i>Abudefduf</i>	<i>saxatilis</i>	2–16	16–18	18	1–2, 18
2	<i>Amphistichus</i>	<i>argenteus</i>	2–14	14–29	29	1–2, 22
3	<i>Centropyge</i>	<i>flavissimus</i>	2–15	16–17	17	1–2, 17
4	<i>Chaetodon</i>	<i>miliaris</i>	2–15	13–15	15	1–2, 15
5	<i>Cheilinus</i>	<i>fasciatus</i>	2–10	10–12	12	1–2
6	<i>Ctenochaetus</i>	<i>striatus</i>	2–14	15–17	17	1–2
7	<i>Gomphosus</i>	<i>varius</i>	2–14	14–16	16	1–2
8	<i>Parupeneus</i>	<i>pleurostigma</i>	2–14	10–17	17	1–2
9	<i>Scarus</i>	<i>frenatus</i>	2–11	11–14	14	1–2
10	<i>Stethojulis</i>	<i>trilineata</i>	2–11	12–13	13	1–2
11	<i>Thorichthys</i>	<i>meeki</i>	2–13	12–14	14	1–3, 12–14
12	<i>Zanclus</i>	<i>cornutus</i>	2–16	17–18	18	1–2, 18

PCA ID, principal components analysis ID number; ADP, adductor profundus; ADR, adductor radialis; NB rays, nonbranched rays. Fin rays are largely in agreement with the literature (Burgess, 1978; Thresher, 1984; Randall et al., 1990).

toral fins appear to share the same basic mechanical design across many taxa, morphological trends associated with the rowing-flapping continuum include low aspect ratio (AR) fins for rowing and high AR fins for flapping, and a more vertical angle of insertion of the pectoral fin onto the body in rowing specialists. We predict that flapping species should have powerful leading edge control through larger arrector ventralis and dorsalis muscles, a higher mechanical advantage due to basal processes on the fin rays, relatively larger muscles that power the downstroke, and radials that move as a unit. Rowers, on the other hand, are expected to have muscles specialized for increased independent control of multiple fin rays, a lower mechanical advantage of fin rays, radials that allow flexibility of fin motion, and larger muscles to power the upstroke.

To test these predictions, this study presents a detailed survey of the pectoral fin musculoskeletal system of 12 species of fishes that use labriform propulsion as the primary locomotor mode over a wide range of speeds. Musculoskeletal diversity and its functional implications are used to address two main questions: 1) Are the major features of pectoral osteology and musculature similar across phylogenetically distant species that use labriform locomotion? To answer this question we surveyed fin shape, the osteology of the fin base, and origin-insertion patterns of muscle subdivisions to assess homology of these structures across taxa. 2) Can variation in labriform locomotion be attributed to variation in musculoskeletal form? We discuss the functional implications of observed morphological variation in the context of previous research on fin kinematics and electromyography.

MATERIALS AND METHODS

Specimens

Pectoral girdles from 12 species (Table 1) were obtained from museum and field collections to represent pectoral fin diversity in perciform fishes with labriform propulsion. At least two specimens were dissected for muscle mass data in all taxa except *Parupeneus pleurostigma* (n = 1), for which specimen availability

was limited. Material examined included the following species: *Abudefduf saxatilis* (FMNH 61658 [1], 120 mm TL; FMNH 61660 [1], 128 mm TL; FMNH 48155 [1], 150 mm total length, TL), *Amphistichus argenteus* (FMNH 42553 [2], 109–115 mm TL; FMNH 42588 [1], 111 mm TL; LACM-W67-149A [1], 188 mm TL), *Centropyge flavissimus* (FMNH 44099 [2], 71–76 mm TL; FMNH 16287 [1], 122 mm TL), *Chaetodon miliaris* (FMNH 70267 [2], 137–146 mm TL), *Cheilinus fasciatus* (FMNH 111974 [1], 230 mm TL; FMNH 111972 [1], 247 mm TL; FMNH 111973 [1], 260 mm TL), *Ctenochaetus striatus* (FMNH 75894 [2], 166–178 mm TL; FMNH 74045 [1], 190 mm TL), *Gomphosus varius* (FMNH 111975 [1], 198 mm TL; FMNH 111977 [1], 240 mm TL; FMNH 111976 [1], 243 mm TL), *Parupeneus pleurostigma* (FMNH 63635 [1], 174 mm TL), *Scarus frenatus* (FMNH 93-63 [1], 213 mm TL; FMNH 111980 [1], 340 mm TL), *Stethojulis trilineata* (FMNH 110756 [1], 120 mm TL; FMNH 111978 [1], 139 mm TL; FMNH 111979 [1], 148 mm TL), *Thorichthys meeki* (FMNH 44225 [1], 125 mm TL; FMNH 104554 [1], 130 mm TL; FMNH 82601 [1], 133 mm TL), and *Zanclus cornutus* (FMNH 44225 [1], 137 mm TL; FMNH 46078 [2], 124–155 mm TL).

Morphology

Fin shape differences were quantified following Walker and Westneat (2002). Fins were pinned to a foam board with the rays in an expanded (splayed) position and brushed with full-strength formalin to preserve them in this position. Digital images were captured using a Spot RT (Diagnostic Instruments, Sterling Heights, MI) digital camera mounted on a Wild M3Z stereomicroscope or a light table. Digital images were then saved in PhotoShop 4.0 (Adobe Systems, San Jose, CA) and analyzed using NIH Image 1.62 on an Apple Macintosh G3 computer. A custom-designed macro was employed in NIH Image to measure and construct variables that reflect functional aspects of fin shape. Measurements included aspect ratio, tip area, mid area, base area, tip chord, mid chord, and base chord of the fin.

A standard dissecting procedure was employed in which specimens were first skinned and hypaxial muscle overlying the pectoral girdle was removed. The two cleithra were separated and the pectoral girdle freed from the body after separating the post-temporal from the neurocranium. Pectoral muscles were then separated using a blunt probe with the aid of a dissecting microscope. Muscle origination and insertion patterns were documented using a dissecting microscope and drawn using a vector based drawing program on a PC (Micrografx designer 8.0).

Each muscle was removed from its origin and cut along the tendons prior to insertion on the fin rays from the left side of all specimens unless otherwise noted. The muscles were blotted dry on paper towels in a standard procedure and muscle mass was measured using a Mettler H10W scale with a precision of 0.0001 g. Raw values of pectoral girdle musculature masses, measured in grams, are reported in the Appendix. Ratios of muscle masses and

percentages of pectoral musculature were determined following Geerlink (1989). We performed a principal components analysis on the correlation matrix of arcsine transformed percentages of the six muscle measurements: abductor superficialis (ABS), abductor profundus (ABP), arrector ventralis (ARRV), adductor superficialis (ADS), adductor profundus (ADP), and arrector dorsalis (ARRD) using the statistical program JMP v. 3.16. An arcsine transformation was performed because muscle weights are presented as percentages of total pectoral muscle mass and this procedure prevents the variances in muscle weights from being a function of the mean (Sokal and Rohlf, 1995).

Osteological data were taken from cleared and stained specimens using the enzyme technique of Taylor and Van Dyke (1985), with chemical concentrations and reaction times tailored for each batch of specimens. Musculoskeletal terminology follows Winterbottom (1974) and Geerlink (1989). Novel muscle subdivisions were designated as components of previously named muscles with Greek letters (α , β , χ), rather than proposing new muscle names.

RESULTS

Fin Shape Analysis

Fin aspect ratio and other variables were consistent with the findings of Walker and Westneat (2002) and Wainwright et al. (2002) (Table 2). Pectoral fin shape, as expressed by aspect ratio, varied among species from a low of 1.96 in *Cheilinus fasciatus* and 2.30 in *Thorichthys meeki* (both species with a rowing stroke) to a high of 4.08 in *Ctenochaetus striatus* and 4.39 in *Abudefduf saxatilis* (both flapping species). Higher aspect ratio fins typically had a relatively long leading edge and were narrower distally. Flappers were characterized by a higher aspect ratio and a longer leading edge relative to trailing edge (e.g., *A. saxatilis*, *Amphistichus argenteus*, *Chaetodon miliaris*, *C. striatus*, *Gomphosus varius*, *Scarus frenatus*, *Stethojulis trilineata*, *Zanclus cornutus*). The distribution of fin area was different among species. The smallest fin tip area and middle area were found predominantly in wrasses. Fin base area was largest in flappers and smallest in rowers, indicating that the center of fin area is located closer to the fin base in flappers.

Muscle Structure

The muscles of the pectoral girdle are composed of abductors on the lateral side and adductors on the medial side, all of which originate on the pectoral girdle. All species surveyed in this study possess the six major pectoral muscle groups of many teleosts: abductor superficialis, arrector ventralis, abductor profundus, adductor superficialis, arrector dorsalis, and adductor profundus (Winterbottom, 1974; Geerlink, 1989). Modifications to this basic muscle plan in muscle size, muscle mass, subdivision of individual muscle components, and variation in orientation of origin and insertion are described below.

Abductor superficialis (ABS). The ABS is the most superficial abductor muscle (Fig. 1A–F). It originates along the anterolateral surface of the cle-

TABLE 2. Fin shape data for the 12 species studied

N	Genus	Species	TL	AR (2 $\times$)	LE span	Total Area	Tip area	Mid area	Base area	Tipchord	Midchord	Basechord
1	<i>Abudefduf</i>	<i>saxatilis</i>	150	4.390	3.159	4.617	0.411	0.955	1.033	0.969	1.408	1.47
1	<i>Amphistichus</i>	<i>argenteus</i>	109	3.864	1.9839	2.037	0.220	0.378	0.446	0.768	1.139	1.012
1	<i>Centropyge</i>	<i>flavissimus</i>	122	2.500	2.048	3.675	0.459	0.693	0.724	1.322	1.723	1.663
1	<i>Chaetodon</i>	<i>miliaris</i>	137	3.478	2.332	3.281	0.356	0.681	0.648	1.183	1.602	0.745
9	<i>Cheilinus</i>	<i>fasciatus</i>	203.22 $\pm$ 66.61	1.957 $\pm$ 0.188	2.862 $\pm$ 1.087	9.099 $\pm$ 5.415	0.788 $\pm$ 0.373	1.560 $\pm$ 0.953	1.685 $\pm$ 1.062	1.974 $\pm$ 0.508	2.778 $\pm$ 0.934	2.602 $\pm$ 0.929
1	<i>Ctenochaetus</i>	<i>striatus</i>	166	4.084	3.4786	5.92732	0.516	1.217	1.311	1.183	2.024	1.766
19	<i>Gomphosus</i>	<i>varius</i>	179.42 $\pm$ 38.33	3.347 $\pm$ 0.397	2.699 $\pm$ 0.609	4.453 $\pm$ 1.592	0.267 $\pm$ 0.087	0.771 $\pm$ 0.243	1.008 $\pm$ 0.395	0.918 $\pm$ 0.159	1.782 $\pm$ 0.258	1.790 $\pm$ 0.347
1	<i>Parupeneus</i>	<i>pleurostigma</i>	174	2.548	2.2382	3.93341	0.590	0.704	0.532	1.474	1.701	0.8803
3	<i>Scarus</i>	<i>frenatus</i>	291.67 $\pm$ 52.99	2.818 $\pm$ 0.399	4.883 $\pm$ 0.679	17.395 $\pm$ 5.468	1.442 $\pm$ 0.71	3.007 $\pm$ 0.891	3.502 $\pm$ 0.926	2.226 $\pm$ 0.545	3.665 $\pm$ 0.587	3.497 $\pm$ 0.502
6	<i>Stethojulis</i>	<i>trilineata</i>	124.50 $\pm$ 22.42	3.762 $\pm$ 0.395	2.085 $\pm$ 0.347	2.406 $\pm$ 0.895	0.128 $\pm$ 0.038	0.376 $\pm$ 0.105	0.566 $\pm$ 0.217	0.610 $\pm$ 0.106	1.232 $\pm$ 0.215	1.359 $\pm$ 0.520
1	<i>Thorichthys</i>	<i>meeki</i>	125	2.302	2.1044	3.84769	0.505	0.706	0.650	1.3683	1.737	1.3301
3	<i>Zanclus</i>	<i>cornutus</i>	82.33 $\pm$ 5.723	3.873 $\pm$ 0.4756	1.757 $\pm$ 0.201	1.63 $\pm$ 0.274	0.154 $\pm$ 0.057	0.254 $\pm$ 0.513	0.349 $\pm$ 0.658	0.615 $\pm$ 0.015	0.861 $\pm$ 0.073	1.121 $\pm$ 0.459

Table entries are means per species. Data from species with more than three as the sample size include additional data from Wainwright et al. (2002).

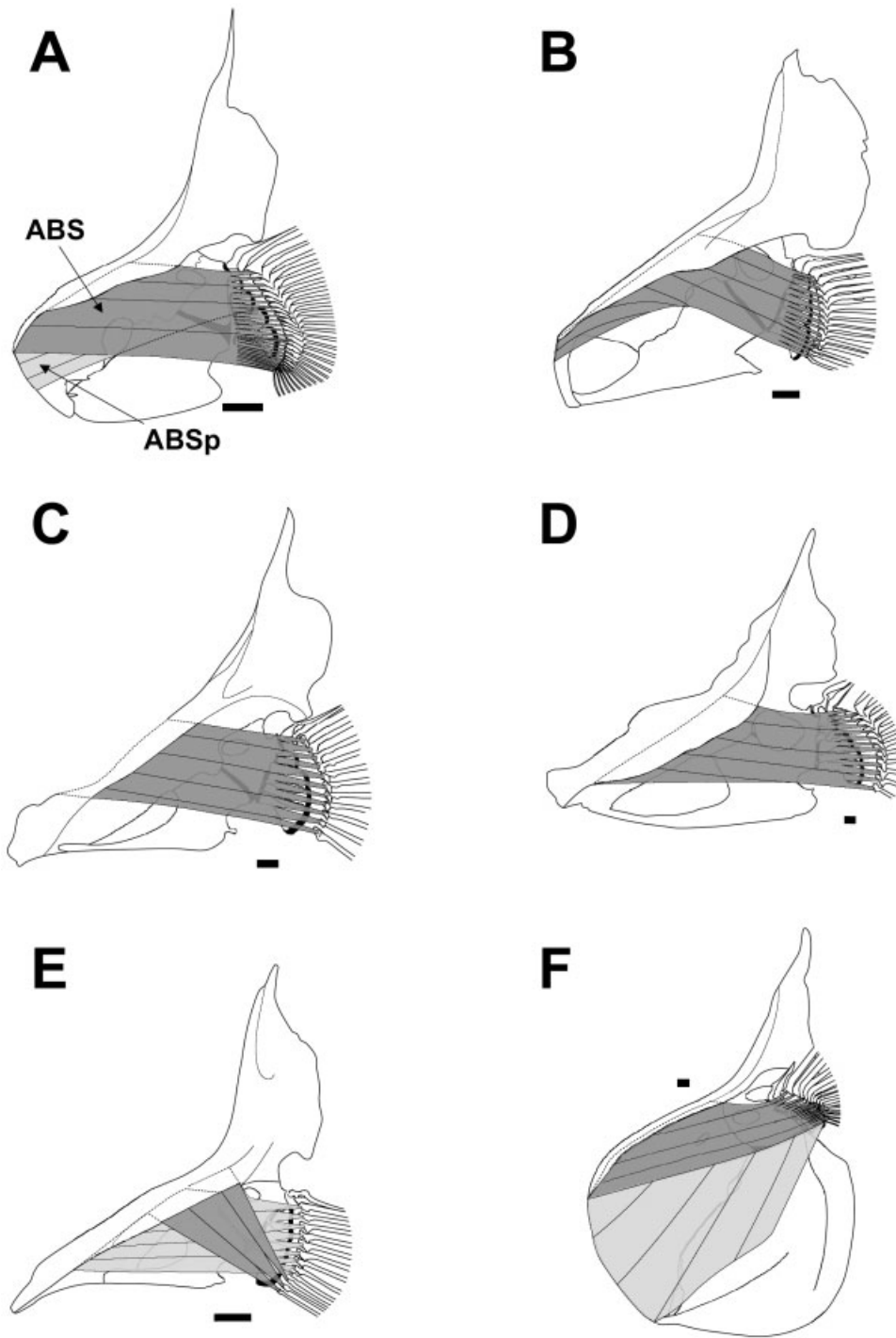


Fig. 1. Lateral views of the pectoral girdle abductor superficialis (ABS) diversity including the superficialis pars superficialis (ABSs) and pars profunda (ABSp) subdivisions. **A:** *Amphistichus argenteus*. **B:** *Chaetodon miliaris*. **C:** *Cheilinus fasciatus*. **D:** *Scarus frenatus*. **E:** *Thorichthys meeki*. **F:** *Zanclus cornutus*. Dark gray indicates ABS and light gray indicates subdivision. Dotted lines indicate muscle margins. Note the presence of muscle subdivisions in **A**, **E**, and **F**. Scale bar = 3 mm.

ithrum and inserts via tendons on processes near the base of all lateral fin rays except the stout first (P1) ray. Novel subdivisions were found in three species (*Amphistichus argenteus*, *Thorichthys meeki*, and *Zanclus cornutus*) that appear to be involved in fin control. It is unclear if these subdivisions are homologous. We refer to them as the abductor superficialis pars superficialis (ABSs) and pars profunda (ABSp) following Geerlink (1979). In

A. argenteus the ABSs inserts onto all fin rays except P1 while the ABSp originates ventrally to the ABSs subdivision and inserts on rays 2–8 (Fig. 1A). In *T. meeki* the ABSs inserts on rays 11–13, while the ABSp originates ventrally to the ABSs subdivision and inserts on rays 2–10 (Fig. 1E). Geerlink (1979) found the ABSs and ABSp to insert on rays 8–11 and 2–8, respectively, in *Sarotherodon niloticus*. In *Z. cornutus* the ABSs and ABSp insert onto all fin

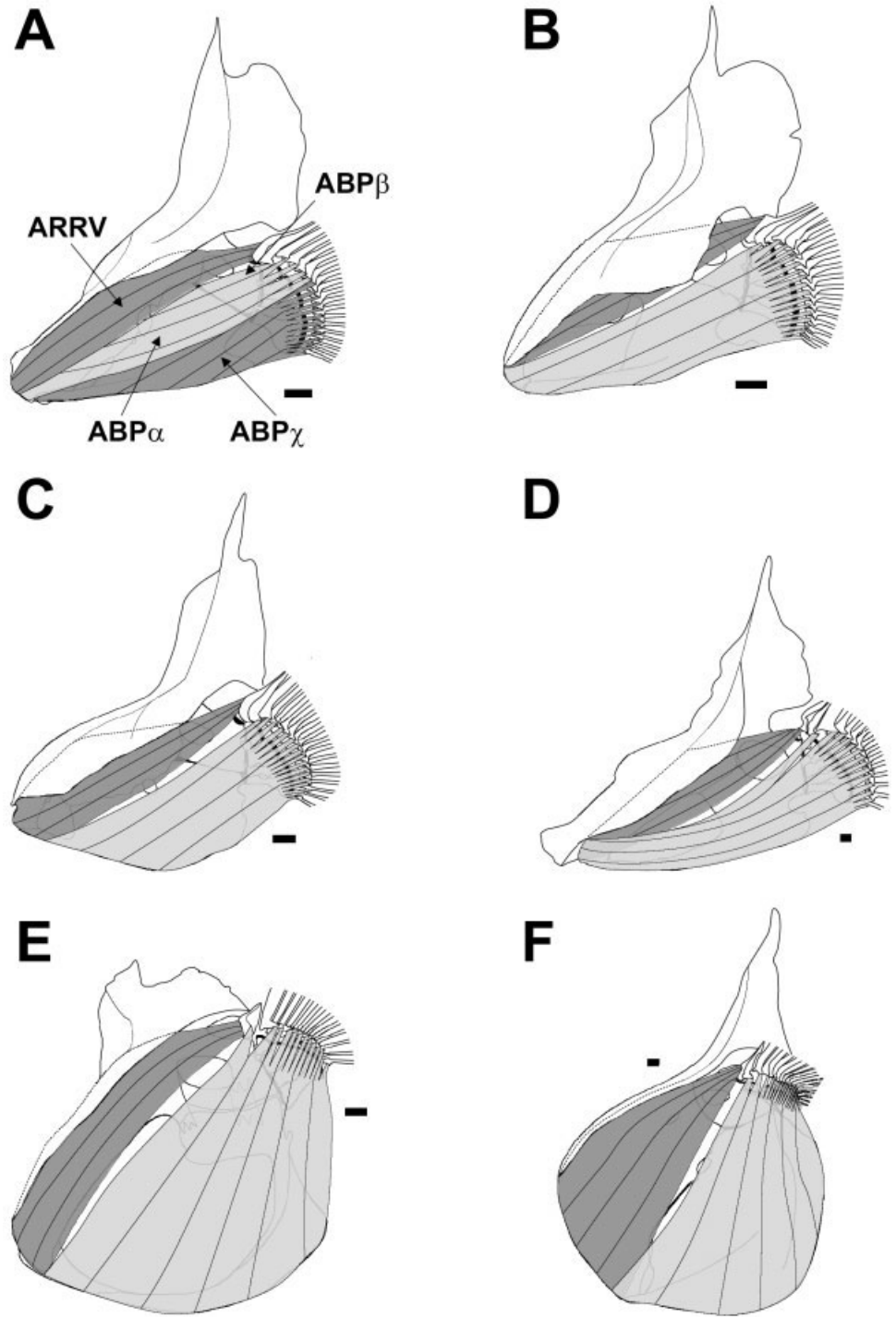


Fig. 2. Lateral views of the pectoral girdle abductor profundus (ABP) diversity including the main subdivision of the abductor profundus ($ABP\alpha$), subdivision to the first pectoral fin ray ($ABP\beta$), and a novel subdivision in *Abudegduf saxatilis* ($ABP\chi$) that originates at the ventral margin of the cleithrum and coracoid along the main body of the $ABP\alpha$ and is located laterally to the ABP and arrector ventralis diversity (ARR). **A:** *Abudegduf saxatilis*. **B:** *Centropyge flavissimus*. **C:** *Ctenochaetus striatus*. **D:** *Scarus frenatus*. **E:** *Stethojulis trilineata*. **F:** *Zanclus cornutus*. Note the presence of a muscle subdivision in **A**. Scale bar = 3 mm.

rays except the first and the ABSp originates ventrally to the ABSs subdivision (Fig. 1F).

Additional variations among taxa include the ABS originating parallel to the cleithrum before altering to a parallel trajectory to the fin rays in *Centropyge flavissimus*, *Chaetodon miliaris* (Fig. 1B), and *Ctenochaetus striatus*. Furthermore, *Cheilinus fasciatus*, *Ctenochaetus striatus*, *Gomphosus varius*, *Parupeneus pleurostigma*, and *Scarus frenatus* all

possess an ABS with parallel trajectories of the posterior tendons; however, the anteroventral portion of the muscle diverges slightly (Fig. 1C,D).

Abductor profundus (ABP). The ABP is located medially to the ABS, lies beside and posterior to the arrector ventralis, and originates on the anteroventral tip of the cleithrum and along the dorsal edge of the coracoid (Fig. 2A–F). It inserts via tendons onto processes near the base of all lateral fin rays. A

well-developed muscle subdivision of the ABP inserts onto the lateral process of P1 in all species. We define the main mass of the abductor profundus as ABP α and the subdivision to P1 as ABP β . Winterbottom (1974) noted that this subdivision to the first ray is lost in some taxa and well developed in others. A second novel subdivision was found in *Abudefduf saxatilis* (ABP χ) that originates at the ventral margin of the cleithrum and coracoid along the main body of the ABP α and is located laterally to the ABP (Fig. 2A). The ABP α inserts on all rays and the ABP χ inserts on the last 13 fin rays. Diversity in the abductor profundus muscle across taxa is largely attributed to variation in shape associated with muscle origination on the varied shapes of the cleithrum and coracoid.

Arrector ventralis (ARRV). The ARRV is located medially to the ABS and originates on the anterolateral surface of the cleithrum and the lateral surfaces of the cleithrum and scapula (Fig. 2A–F). The ARRV was one of the most conservative structures examined, consisting of one large muscle bundle that inserts onto the anteromedial base of the P1 ray in all species examined. Orientation of the muscle bundle is similar throughout all species, although there was diversity in the width of the origin.

Adductor superficialis (ADS). The most medial adductor muscle is the ADS (Figs. 3, 4). It originates on the dorsal half of the anteromedial surface of the cleithrum and inserts via tendons onto the more distal processes of the medial sides of all rays except the first. The muscle is folded, with fibers to the leading edge rays lying medially to and making almost right angles with those to the trailing edge rays (Winterbottom, 1974; Geerlink, 1979, 1989). The medial fold inserts on rays 2–4 and the lateral fold inserts on the remaining fin rays.

Adductor medialis (ADM). The adductor medialis (ADM) has an erratic occurrence in teleosts (Winterbottom, 1974) and among the species studied here. The ADM was present in *Centropyge flavissimus*, *Chaetodon miliaris*, *Ctenochaetus striatus*, and *Parupeneus pleurostigma* (Fig. 3B,C), but absent from other taxa. The ADM lies between the ADS lateral and medial folds and inserts onto the middle fin rays, joining with the ADS lateral tendons.

Adductor profundus (ADP). The ADP lies lateral and ventral to the superficialis (Fig. 4). It originates on the anteroventral corner of the cleithrum and along the medial surface of the coracoid and inserts via tendons onto processes near the base of the lateral sides of most fin rays. The ADP exhibits striking interspecific differences in insertion pattern (Table 1). *Ctenochaetus striatus* is the only species in which the ADP inserts on all fin rays excluding P1. On the other hand the insertion of the ADP in *Amphistichus argenteus* ends 16 rays from the trailing edge. The majority of species possess insertions ending three rays from the trailing edge (e.g., *Abudefduf saxatilis*, *Cheilinus fasciatus*, *Centropyge flavissi-*

mus, *Gomphosus varius*, *Stethojulis trilineata*, and *Zanclus cornutus*). Another type of variation is the site of origin on the cleithrum and coracoid.

Arrector dorsalis (ARRD). Beneath the more dorsally originating bundles of the adductor superficialis and just beside the more ventrally originating bundles lies the ARRD (Fig. 4A,F). This muscle originates on the anteromedial surface of the cleithrum, ventral to the origin of the ADS, and inserts via a short tendon onto the base of the first medial fin ray. Orientation of the muscle bundle is similar throughout all species.

Adductor radialis (ADR). The ADR is a small muscle that is located medial to all other adductor muscles, with small bundles converging anteriorly. The ADR originates mainly on the coracoid with slight variation throughout all species. Most species have at least one bundle originating on the scapula. The insertion pattern exhibits striking differences among species (Table 1). In some cases the ADP and ADR insert to a common fin ray (e.g., *Abudefduf saxatilis*, *Amphistichus argenteus*, *Chaetodon miliaris*, *Cheilinus fasciatus*, *Gomphosus varius*, *Parupeneus pleurostigma*, *Scarus frenatus*, and *Thorichthys meeki*) while in other cases they insert on independent rays (e.g., *Ctenochaetus striatus*, *Centropyge flavissimus*, *Stethojulis trilineata*, and *Zanclus cornutus*).

Muscle Mass Analysis

Muscle masses were determined to estimate the force capacity of the abductor and adductor muscles across species (Table 3). The pectoral fin musculature comprises 0.7–8.2% of the total body mass in the 12 species. Two species, *Stethojulis trilineata* and *Zanclus cornutus*, have the largest pectoral musculature, 6.8% and 8.2%, respectively. The majority of species ranged from 0.3–1.2%. The ABD/ADD indicates the relative mass of the abductors and adductors and ranged from 0.72–1.46. All labrid fishes measured had values <1.0. Three species within the family Labridae and Scaridae with high-performance flapping pectoral fin locomotion (*Gomphosus varius*, *Scarus frenatus*, and *S. trilineata*) have a combined arrector musculature (ARRV + ARRD) that is almost twice as large as many of the other nine species (Table 3, Fig. 5). Additional muscle mass variation in the 12 species is depicted in Figure 5.

The principal components analysis of the ratios of six muscle weights to body mass yielded three components that accounted for 95.8% of the variance in the dataset (Fig. 6, Table 4). PC1 explains much of the variation in the ABS, ADS, and ARRD; PC2 explains variation in the ADP and ABP; PC3 explains variation in the ARRD and ADS (Fig. 6). Wrasses are characterized as having a medium to large ADP and ABP (PC2), ARRD, ADP, and a medium to small ABS (PC1) compared to the other species. This trend in muscle masses remains strong

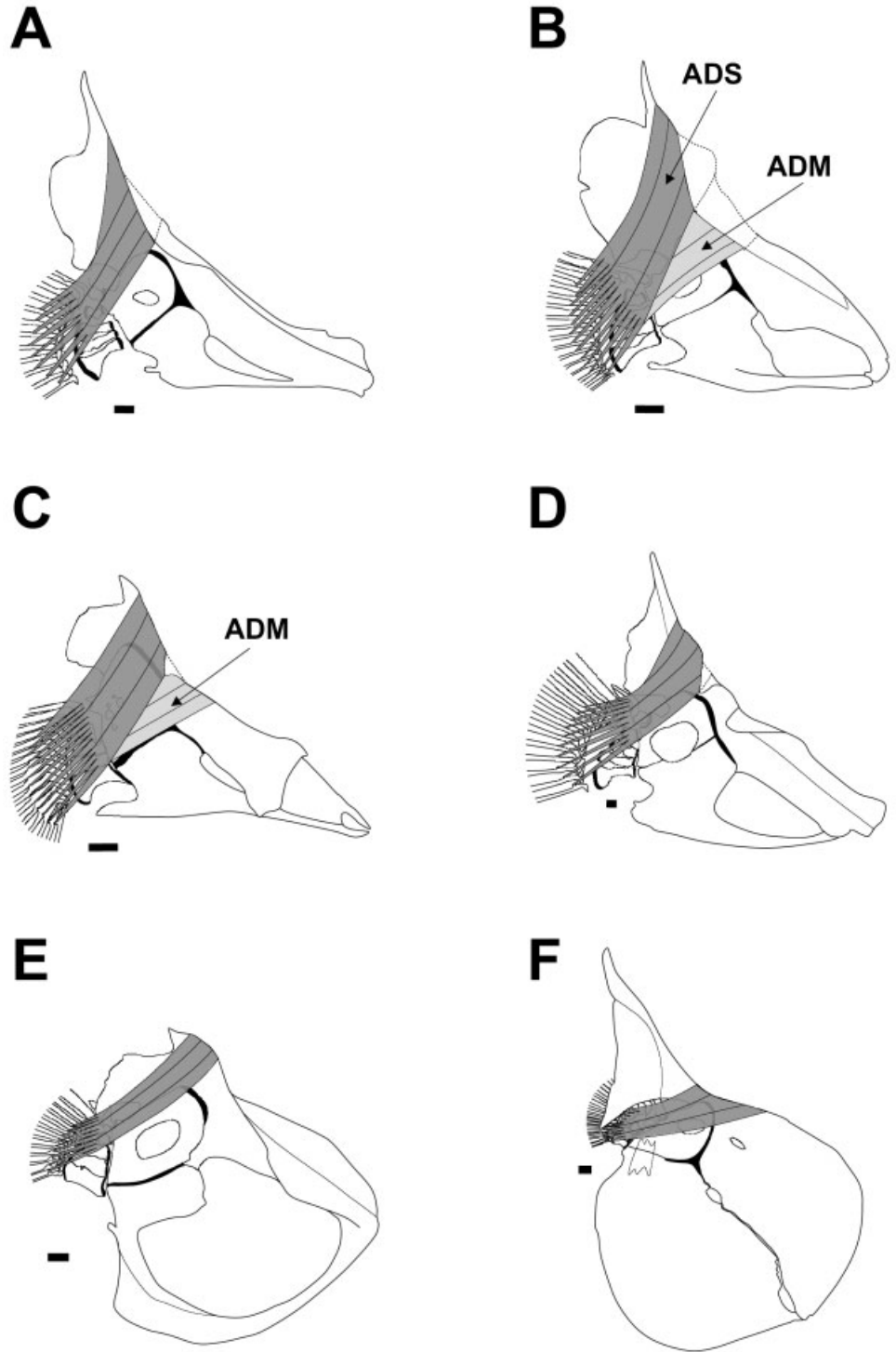


Fig. 3. Medial views of the pectoral girdle adductor superficialis (ADS) and adductor medialis (ADM) diversity. The ADM lies between the ADS lateral and medial folds and inserts onto the middle fin rays, joining up with the ADS lateral tendons. **A:** *Cheilinus fasciatus*. **B:** *Centropyge flavissimus*. **C:** *Parupeneus pleurostigma*. **D:** *Scarus frenatus*. **E:** *Stethojulis trilineata*. **F:** *Zanclus cornutus*. Note the presence of the ADM in **B, C**. Scale bar = 3 mm.

in a larger dataset of 21 additional wrasse species (Thorsen and Westneat, unpubl. data).

Skeletal Structure

Cleithrum, coracoid, and scapula. The scapula and coracoid are attached to the medial lamina of the ventral half of the cleithrum. The ventral tip

of the coracoid does not contact the cleithrum in *Gomphosus varius*. Differences in shape between the 12 species are noticeable in the coracoid, dorsal, and ventral flange of the cleithrum and foramen in the scapula (Figs. 7, 8). The most unusual pectoral girdle is *Zanclus cornutus*, in which the cleithrum and coracoid are expanded and constitute the an-

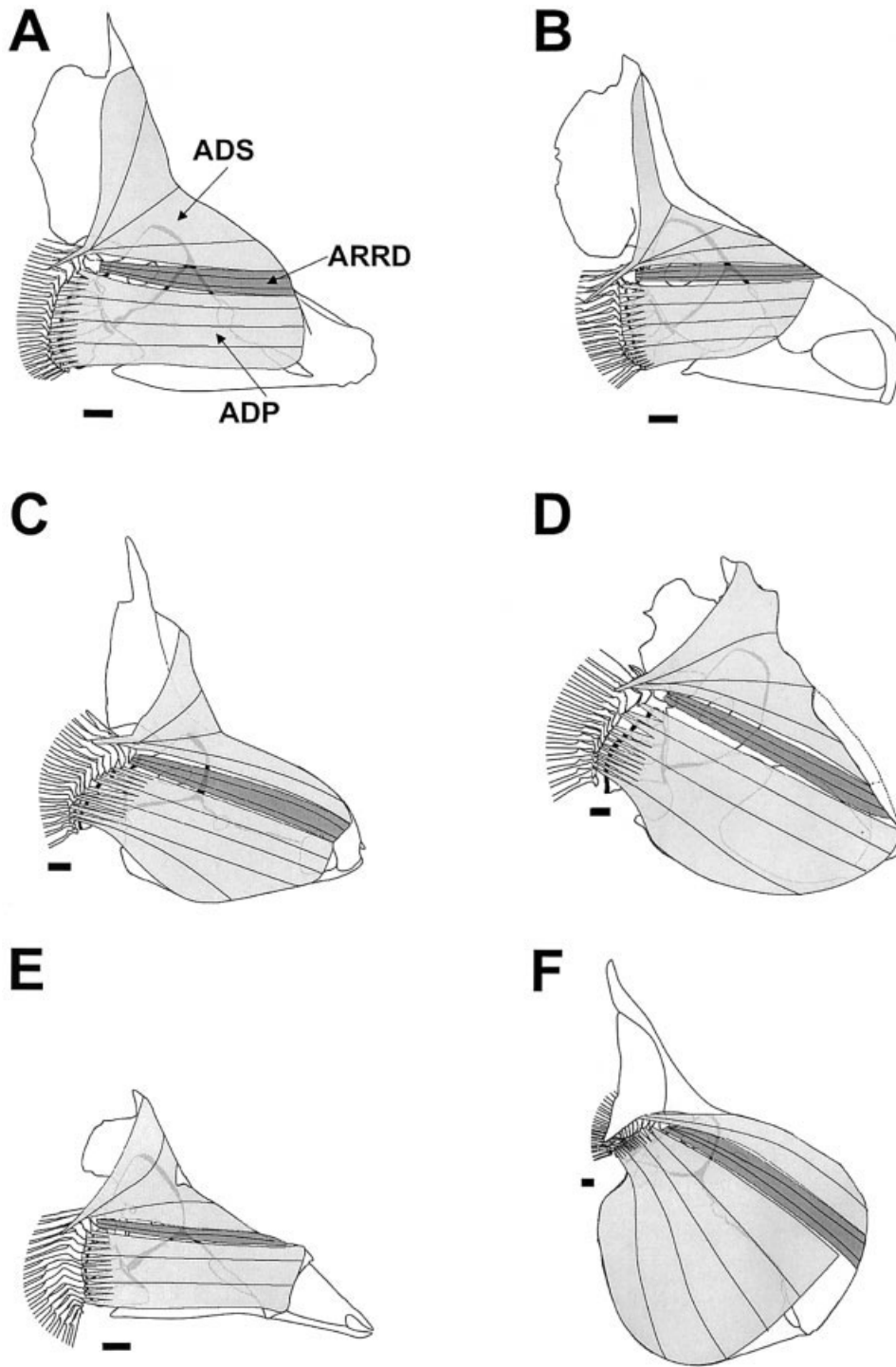


Fig. 4. Medial views of the pectoral girdle adductor profundus (ADP), adductor superficialis (ADS), and arrector dorsalis (ARRD) diversity. **A:** *Abudegduf saxatilis*. **B:** *Chaetodon miliaris*. **C:** *Ctenochaetus striatus*. **D:** *Gomphosus varius*. **E:** *Parupeneus pleurostigma*. **F:** *Zanclus cornutus*. Scale bar = 3 mm.

teroverventral profile of the body between the snout and anal fin. The scapula in *Parupeneus pleurostigma* possesses numerous foramina for nerves, while the other species have a single opening.

Fin rays and processes. Each ray is composed of two hemitrichia (half rays) that are either branched or unbranched (Table 1). Generally, flapping and rowing fins have the first two rays unbranched. In the extreme case in *Thorichthys meeki*, the first

three and last three rays do not branch. Wrasses have all trailing edge branched rays. The majority of other taxa possess at least one trailing edge unbranched fin ray. Each hemitrichium has one or more processes near its base on which the tendons of the abductor or adductor muscles insert. Processes are more pronounced toward the leading edge and gradually become smaller toward the trailing edge (Fig. 9A–F). The most striking differences are found

TABLE 3. Pectoral girdle musculature masses from the left side of specimens

Genus	Species	N	TL	BM (g)	GM (g)	GM/BM %	ABD/ ADD	ARRV/ ARRD	ARRV + ARRD %	ABS/GM %	ABP/ GM%	ARRV/ GM%	ADS/ GM%	ADP/ GM%	ARRD/ GM%
<i>Abudefduf</i>	<i>saxatilis</i>	3	132.67 ± 15.53	43.87 ± 12.05	.286 ± .013	0.690 ± 0.219	1.105 ± 0.084	1.977 ± 0.277	12.101 ± 0.848	19.341 ± 1.303	25.087 ± 0.400	8.010 ± 0.579	21.621 ± 0.466	21.850 ± 1.852	4.091 ± 0.512
<i>Amphistichus</i>	<i>argenteus</i>	4	130.75 ± 38.25	36.50 ± 39.96	.315 ± .326	0.906 ± 0.106	1.134 ± 0.131	1.751 ± 0.189	14.456 ± 1.374	16.094 ± 1.491	27.743 ± 1.995	9.166 ± 0.6780	15.504 ± 2.923	26.203 ± 4.165	5.290 ± 0.764
<i>Centropyge</i>	<i>flavissimus</i>	3	89.67 ± 28.11	22.96 ± 22.70	.126 ± .119	0.568 ± 0.035	1.460 ± 0.158	2.790 ± 0.868	10.570 ± 0.426	23.644 ± 0.788	27.947 ± 2.201	7.650 ± 0.603	18.202 ± 2.457	19.637 ± 1.774	2.921 ± 0.863
<i>Chaetodon</i>	<i>miliaris</i>	2	141.50 ± 6.36	48.53 ± 10.32	.209 ± .035	0.433 ± 0.021	1.459 ± 0.067	2.112 ± 0.539	10.557 ± 0.918	29.403 ± 1.297	22.769 ± 1.190	7.139 ± 1.215	18.187 ± 1.767	19.084 ± 0.956	3.418 ± 0.297
<i>Cheilinus</i>	<i>fasciatus</i>	3*	245.67 ± 15.04	311.90 ± 109.32	1.009 ± .342	1.009 ± 0.342	0.868 ± 0.097	1.537 ± 0.053	11.378 ± 0.544	15.763 ± 0.558	23.711 ± 2.689	6.895 ± 0.423	20.363 ± 1.053	28.785 ± 3.095	4.4829 ± 0.121
<i>Ctenochaetus</i>	<i>striatus</i>	3	178.00 ± 12.00	80.27 ± 11.74	.935 ± .047	1.175 ± 0.112	1.197 ± 0.019	1.736 ± 0.202	13.598 ± 0.577	19.791 ± 1.419	26.077 ± 0.818	8.618 ± 0.677	21.931 ± 0.946	18.603 ± 1.096	4.980 ± 0.243
<i>Gomphosus</i>	<i>varius</i>	3	227.00 ± 25.16	108.53 ± 28.20	1.377 ± .642	1.216 ± 0.332	0.717 ± 0.069	0.870 ± 0.069	17.560 ± 0.903	13.057 ± 2.048	20.455 ± 1.279	8.168 ± 0.083	16.222 ± 0.807	32.666 ± 1.729	9.432 ± 0.832
<i>Parupeneus</i>	<i>pleurostigma</i>	1**	174.00	52.88	.349	0.659	0.840	1.423	9.355	19.018	21.156	5.474	20.810	29.661	3.881
<i>Scarus</i>	<i>frenatus</i>	2	276.50 ± 89.80	511.75 ± 490.10	3.620 ± 3.166	0.759 ± 0.109	0.834 ± 0.041	1.034 ± 0.056	18.486 ± 2.277	12.133 ± 0.194	23.942 ± 0.518	9.378 ± 0.907	18.543 ± 0.282	26.897 ± 2.883	9.1080 ± 1.370
<i>Stethojulis</i>	<i>trilineata</i>	3	135.67 ± 14.29	38.14 ± 13.85	1.293 ± .450	3.422 ± .237	0.934 ± 0.032	1.094 ± 0.017	19.636 ± 0.587	7.061 ± 0.648	30.972 ± 1.111	10.261 ± 0.374	12.122 ± 1.187	30.209 ± 1.041	9.375 ± 0.219
<i>Thorichthys</i>	<i>meeki</i>	3	129.33 ± 4.04	30.24 ± 7.33	.100 ± .026	0.337 ± 0.081	0.898 ± 0.053	2.618 ± 0.172	8.324 ± 0.801	26.719 ± 1.207	14.527 ± 1.845	6.025 ± 0.666	19.437 ± 1.275	30.992 ± 2.005	2.300 ± 0.161
<i>Zanclus</i>	<i>cornutus</i>	3	138.67 ± 15.57	59.64 ± 31.64	2.891 ± 2.304	4.121 ± 2.360	1.191 ± 0.085	1.945 ± 0.339	17.343 ± 0.938	11.419 ± 1.417	31.516 ± 1.405	11.377 ± 0.277	10.596 ± 2.103	29.126 ± 2.913	5.965 ± 0.996

ABS, abductor superficialis; ABP, abductor profundus; ARRV, arrector ventralis; ADS, adductor superficialis; ADP, arrector dorsalis. All are in % of pectoral muscle mass. Body mass (BM) and girdle musculature (GM) are provided to calculate individual muscle masses. GM/BM%, girdle musculature/body mass *100. N, number of specimens examined.

*One specimen excluded from BM because stomach contents were removed.

**Left and right pectoral fins were included in the analysis.

Data were recorded to 0.0001 g, but are only presented in this table as 0.001 g.

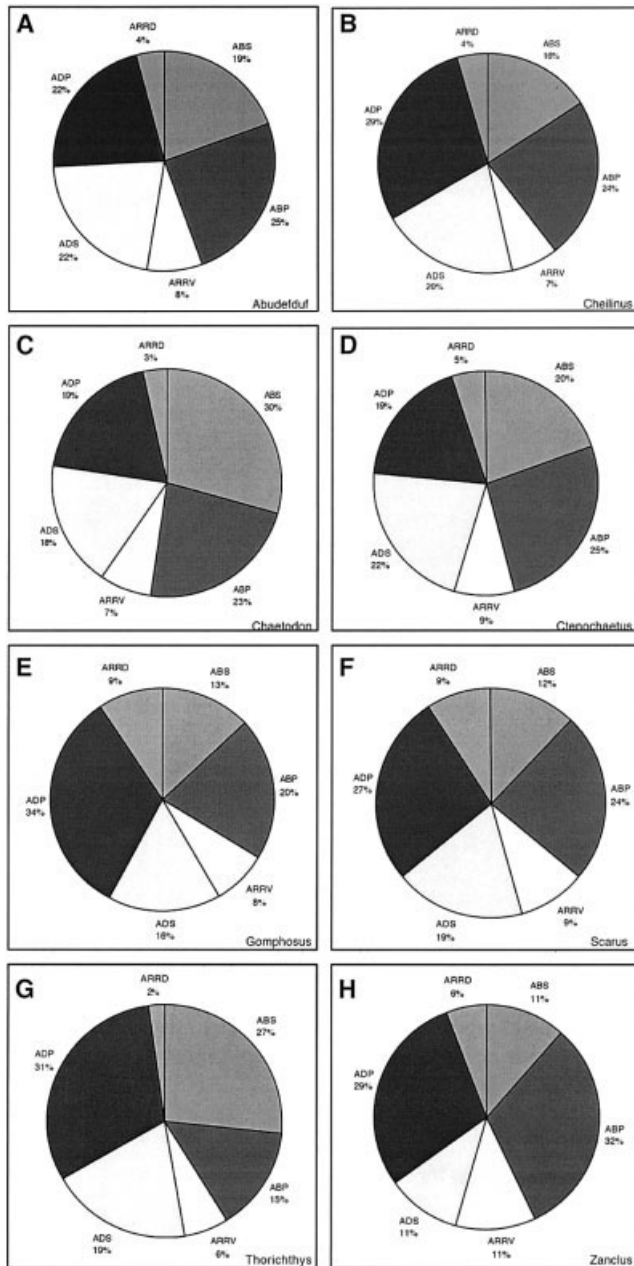
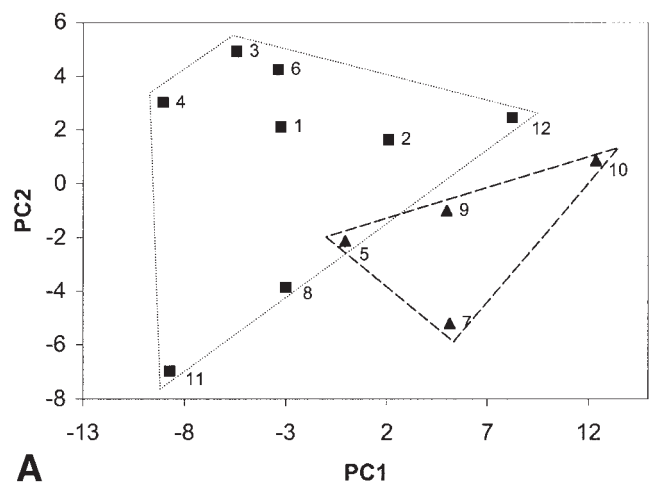


Fig. 5. Muscle mass percentages for each of the six major pectoral muscles in eight of the species studied including the abductor superficialis (ABS), abductor profundus (ABP), arrector ventralis (ARRV), adductor superficialis (ADS), adductor profundus (ADP), and arrector dorsalis (ARRD). **A:** *Abudedefduf saxatilis*. **B:** *Cheilinus fasciatus*. **C:** *Chaetodon miliaris*. **D:** *Ctenochaetus striatus*. **E:** *Gomphosus varius*. **F:** *Scarus frenatus*. **G:** *Thorichthys meeki*. **H:** *Zanclus cornutus*.

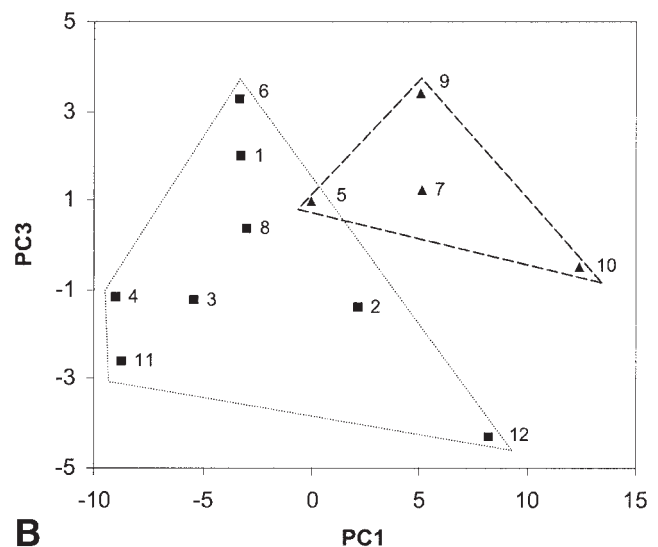
between *Cheilinus fasciatus* and *Gomphosus varius*. In *C. fasciatus* the ABS and ABP have individual processes, each of which is on opposite sides of the point of rotation from middle to trailing edge rays. Actuation on an individual tendon yields extensive control over a single ray. *Gomphosus varius* possesses a process that extends in the direction of the

trailing rays and becomes smaller from the leading edge to the trailing edge. In addition, the ABS and ABP insert to the right of the point of rotation in this species. In this species, actuation of an individual tendon yields little control for a single ray.

Radials and intraradial ligaments. Proximal radials of the 12 species are composed of four bony elements that articulate with the coracoid and scapula. Their size increases toward the trailing edge and their distal ends are embedded in a fibrocartilage pad. The fin base articulates with the posterior edge of the pectoral girdle, attaching directly to the scapula anteriorly, and to the cartilage pad at the end of the radials posteriorly. We discovered a set of previously undescribed intraradial ligaments that



A



B

Fig. 6. Principal components analysis. **A:** Scatterplot of species scores on principal component 2 (PC2 an axis of variation in the ADP and ABP) against PC1 (an axis of variation in the ABS, ARRD, and ADS). **B:** Scatterplot of species scores on PC3 (an axis of the ARRD and ADS) vs. PC1. ▲, parrotfish and wrasses; ■, all other reef fish studied herein. See PCA IDs in Table 1.

TABLE 4. Eigenvalues and eigenvector loadings for the principal component analysis of muscle masses of six components

Principal component	1	2	3	4	5	6
Eigen value	45.063	14.624	5.360	2.383	0.435	0.057
Percent.	66.346	21.531	7.891	3.508	0.640	0.084
Cum percent.	66.346	87.877	95.768	99.276	99.920	100
Factor loadings						
ABS	-0.736	0.070	-0.264	0.374	-0.173	0.463
ABP	0.284	0.659	-0.148	-0.384	-0.318	0.463
ARRV	0.206	0.230	-0.080	0.283	0.841	0.334
ADS	-0.309	-0.098	0.758	-0.409	0.173	0.350
ADP	0.267	-0.703	-0.339	-0.236	-0.030	0.512
ARRD	0.410	-0.070	0.461	0.640	-0.363	0.273

connect each radial to one another, radial one to the first ray, and the base of radial one or two to the scapula (Fig. 9). These ligaments are easily visible under polarized light and they appear to function as stabilizers of the radial complex to produce a single rotating unit. In some species the intraradial ligaments are at the posterior margins of the radials (Fig. 9A–D,F), while in others, radials two through four are sutured, not containing ligaments (Fig. 9E). In addition, the number of ligaments connecting the P1 ray to the first radial and ligaments connecting radial two and three to the scapula show considerable diversity (Fig. 9A–F). The connection between the P1 ray and the first radial is composed of one ligamentous connection in all species except in *Centropyge flavissimus*, in which there are two. There is at least one ligament that connects the radials to the scapula. In *Cheilinus fasciatus*, *Chaetodon miliaris*, and *Scarus frenatus* all possess a scapula-to-first radial ligament, while the other species possess a scapula-to-radial two ligament.

DISCUSSION

Many fishes swim over a wide range of speeds using the labriform mode, powering the pectoral fins with a complex system of muscles for both upstroke and downstroke. This morphological study demonstrates that there are numerous interspecific differences in muscle subdivisions, muscle masses, and skeletal morphology that have functional relevance to this mode of locomotion. Particular muscle subdivisions may function to enhance the coordination and control of the fin. Disparity in muscle masses may be attributed to specialized locomotor behaviors or to alternate solutions for thrust production generated during independent evolution of labriform propulsion in different groups. Structural differences are present in the intraradial ligaments, shape of the radials, and structure of the fin base processes that affect fin rotation. Here we use the morphology of the pectoral fin in 12 diverse species to develop hypotheses of mechanical design and locomotor function in fishes that employ pectoral fin propulsion.

Muscle Diversity and Locomotor Function

The pectoral girdle musculature of the species shared a common overall design of abductor and adductor muscles specialized for oscillation of the pectoral fins to generate locomotor thrust. This is not surprising because we chose species that use pectoral fin swimming as a primary mode of locomotion. However, our most striking result is the discovery of undescribed muscle subdivisions and the presence of a myriad of muscle mass configurations that have strong functional implications for fin locomotion. Notable diversity includes the specialized subdivisions of the ABP and ABS, and the presence or absence of the ADM.

The abductor profundus was composed of two or more subdivisions in all species surveyed. The two common subdivisions, ABP α and β , are separated by fascia that extend the length of the subdivisions. The main body of the ABP, subdivision α , provides control to most of the fin rays via its numerous tendons inserting on the fin ray bases. We suggest that the ABP β bundle, along with the ARRV and ARRD, controls the movement of the leading edge of the fin. It is not known whether the ABP subdivisions are activated independently (i.e., different duration and onset time) of the main ABP subdivision to function in leading edge control as well. The main ABP α has been shown to have the longest duration of activity of the three abductor muscles during abduction in *Gomphosus varius* (Westneat and Walker, 1997). Further investigation of motor patterns using EMG data from multiple sites in muscles may elucidate the timing of specific subdivision activity.

The ABS subdivisions in *Amphistichus argenteus*, *Thorichthys meeki*, and *Zanclus cornutus* have different lines of action and therefore we suggest alternative functional roles for these muscles. In *A. argenteus* and *Z. cornutus*, the ABSp subdivision produces the main dorsoventral flapping motion, whereas the ABSs appears to function in fin protraction, rotating the anterior fin rays anteriorly. In *T. meeki* we predict that the ABSp provides the majority of the force for fin abduction while the ABSs pulls on the trailing edge rays during braking and maneuvering.

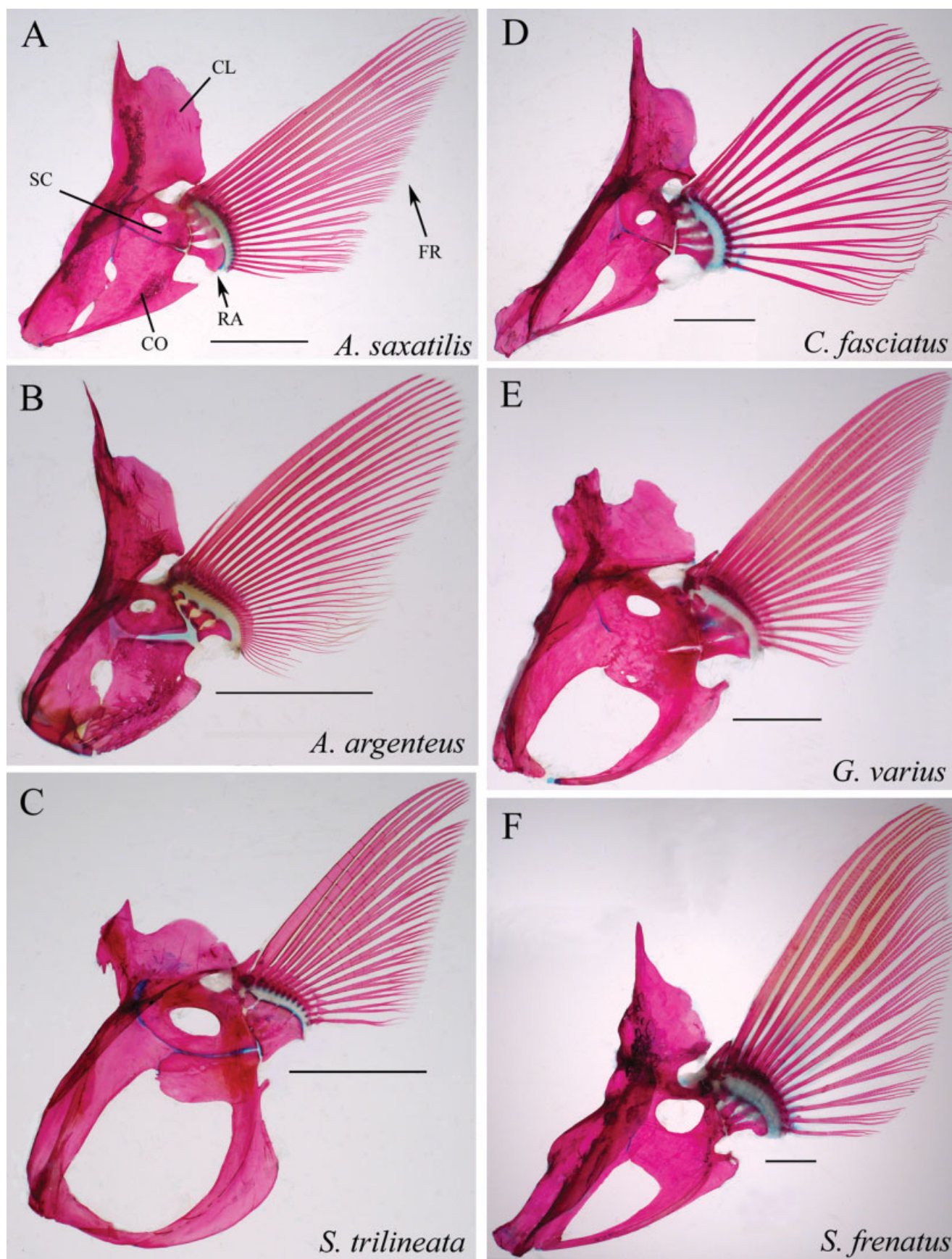


Fig. 7. Skeletal diversity of cleared and stained pectoral girdles and pectoral fin shapes in lateral view of labroid species. **A:** *Abudedefduf saxatilis*. **B:** *Amphistichus argenteus*. **C:** *Stethojulis trilineata*. **D:** *Cheilinus fasciatus*. **E:** *Gomphosus varius*. **F:** *Scarus frenatus*. CL, cleithrum; CO, coracoid; FR, fin rays; RA, radials; SC, scapula. Scale bar = 1 cm.

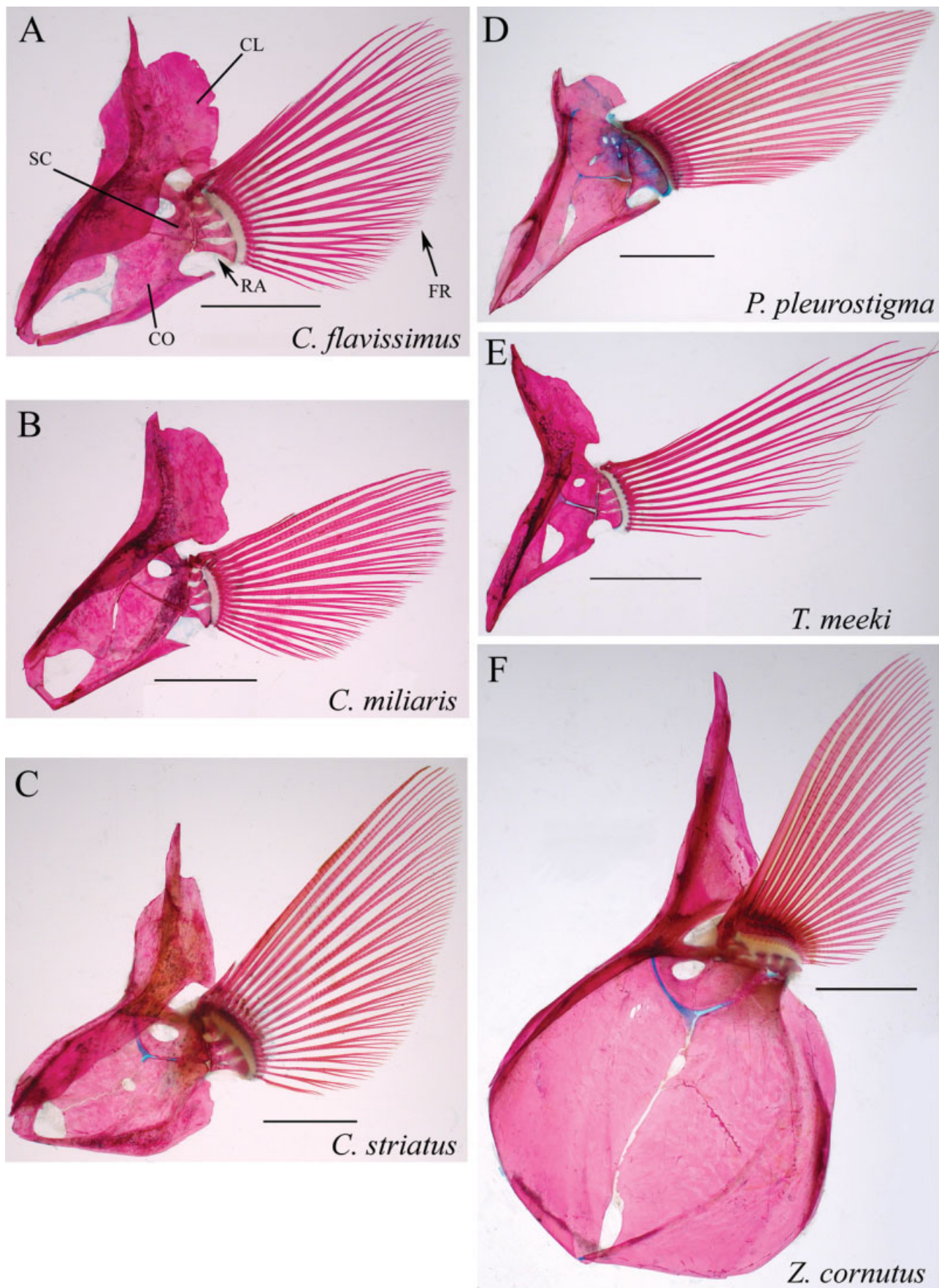


Fig. 8. Skeletal diversity of cleared and stained pectoral girdles and pectoral fin shapes in lateral view of several perciform families. **A:** *Centropyge flavissimus*. **B:** *Chaetodon miliaris*. **C:** *Ctenochaetus striatus*. **D:** *Parupeneus pleurostigma*. **E:** *Thorichthys meeki*. **F:** *Zanclus cornutus*. CL, cleithrum; CO, coracoid; FR, fin rays; RA, radials; SC, scapula. Scale bar = 1 cm.

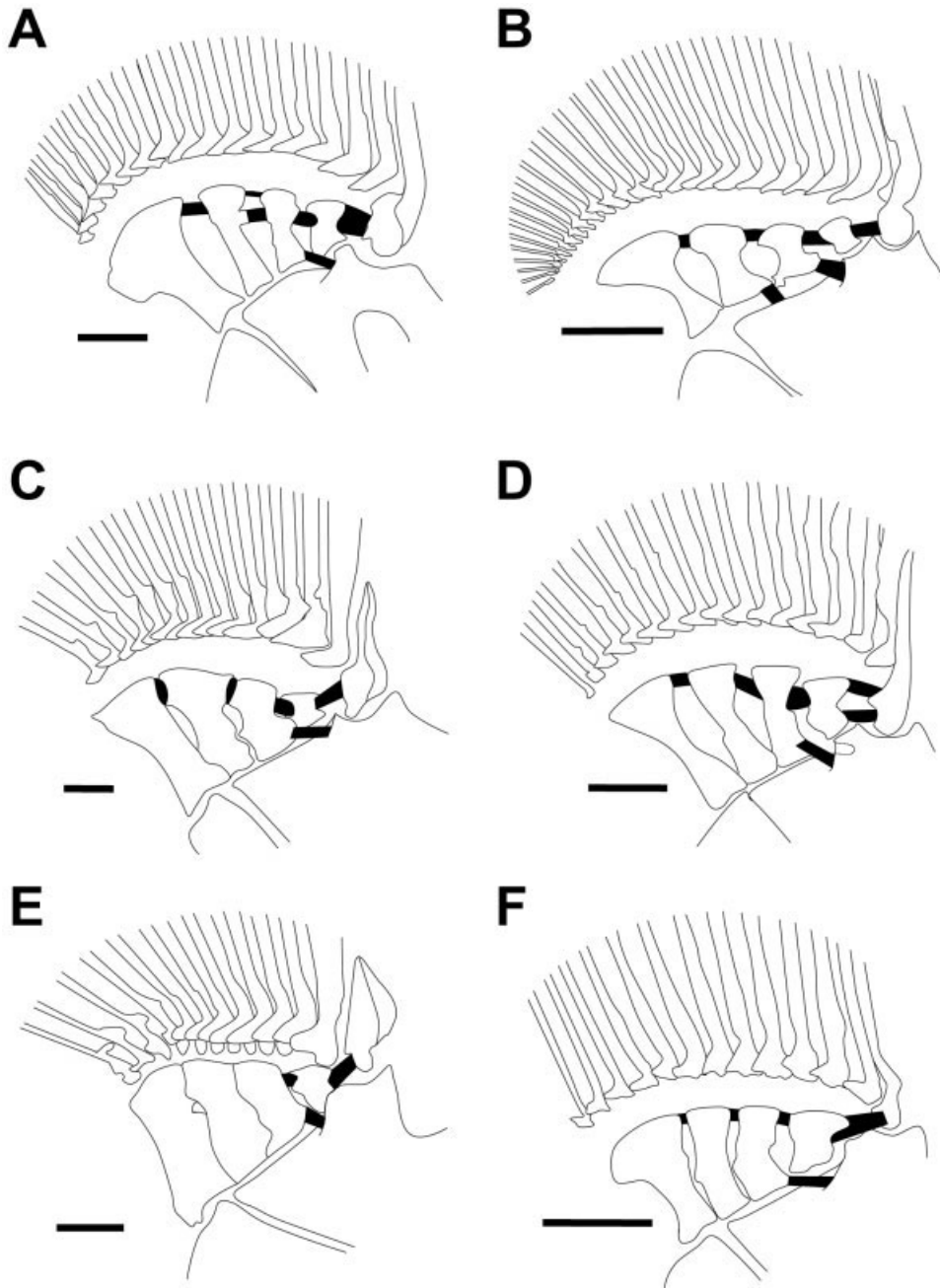


Fig. 9. Radial ligament diversity in medial view, with heavy black lines indicating ligamentous connections between radials and between the radials and the scapula. Leading edge (anterior) is to the right. **A:** *Abudedefduf saxatilis*. **B:** *Amphistichus argenteus*. **C:** *Gomphosus varius*. **D:** *Centropyge flavissimus*. **E:** *Stethojulis trilineata*. **F:** *Thorichthys meeki*.

The adductor medialis was present in *Centropyge flavissimus*, *Chaetodon miliaris*, *Ctenochaetus striatus*, and *Parupeneus pleurostigma*. Its occurrence had previously been reported only in *Lota*, *Perca*, and gobiids (Grenholm, 1923; Eggert, 1929; Harris, 1960). The presence of the ADM in butterflyfishes, surgeonfishes, and goatfishes suggests that the ADM has probably arisen independently in these families of reef fishes. Although no EMG data have been recorded for the ADM, the origin and insertion of this muscle indicate that it provides control of the central and posterior fin rays.

The relatively invariant positions and attachments of the arrector ventralis and arrector dorsalis muscles may result from their dedicated role in control of the hydrodynamically important leading edge of the fin. Patterns of muscle activity during pectoral fin abduction and adduction cycles (Westneat and Walker, 1997) indicate that the downstroke begins with activity of the ARR_V, and the upstroke begins with activity in the ARR_D, just before the nearly synchronous activity of the remaining adductors. The ARR_V protracts the fin and accelerates the initial downstroke, whereas the ARR_D reverses the

direction of the leading edge to initiate the upstroke. The mass of these muscles is highly variable across taxa. Species with flapping pectoral fin locomotion (*Gomphosus varius*, *Scarus frenatus*, and *Stethojulis trilineata*) have a combined arrector musculature (ARRV + ARRD) that is almost twice as large as many of the other nine species (Table 3, Fig. 5). We suggest that larger arrectors are correlated with a more dorsoventral fin stroke and a steeper stroke plane angle of the fin blade during fin oscillation. Stroke plane angle is tightly correlated with the swimming performance and endurance of pectoral fin swimmers across a wide range of speeds (Walker and Westneat, 2002), highlighting the importance of the arrectors in fin function.

The principal components analysis of the muscle mass data (Fig. 6) indicates that fishes with labriform propulsion show trends that may be attributed to specialized locomotor behaviors. Wrasses tend to occupy their own region on the PCA analysis, while the other species displayed more variation. Wrasses are characterized as having a relatively large ADP, ABP, ARRD, and ADP and a relatively small ABS compared to the other taxa. This trend of the Labridae holds true when the sample size is increased to 30 labrid species (Thorsen and Westneat, unpubl. data). The relative mass of abductors and adductors (Table 3) provides an estimate of the total amount of muscle tissue dedicated to downstroke and upstroke power. The three wrasses and the parrotfish all have an abductor/adductor ratio less than one; *Parupeneus pleurostigma* and *Thorichthys meeki* group with the wrasses on PC2, reflecting the fact that these species also have a larger adductor than abductor musculature. All the remaining flappers have a larger abductor musculature. We suggest that these differences highlight the relative importance of the upstroke (for wrasses and others with a more massive set of adductors) or downstroke (for other flappers with high relative mass of abductors) for thrust production. We predict that the morphospace delimited here for muscle masses is a relatively small region of pectoral fin functional morphospace that will expand when species are added that do not use the pectoral fins as a primary mode of locomotion across a wide range of speeds.

Osteological Diversity and Fin Mechanical Design

The biomechanics of fin motion and thrust production depend on the transmission of muscle force and motion to individual fin rays. Force transmission to the fin is determined by the contractile mechanics of the muscles and the structure of the fin skeleton. The osteological differences in fin shape, fin ray morphology, bony processes at the bases of the fin rays, and the fin radials have functional implications. Fin shape has clear consequences for fin thrust mechanics, related to a trade-off between efficiency of fast

swimming and maneuverability in slowly swimming species. Walker and Westneat (2000, 2002) demonstrated that a rowing stroke is capable of producing stronger thrust transients for maneuvers when it is performed with a more rounded, distally broadened, paddle-shaped fin. In contrast, a more slender, tapering, wing-like fin is more efficient for swimming at higher sustained speeds using a dorsoventral flapping stroke. There are several key parameters of fin shape that are associated with fin motion and locomotor strategy, including fin aspect ratio, fin chord lengths, and the distribution of fin area (Walker and Westneat, 2000, 2002).

Aspect ratios of the pectoral fins of the species surveyed ranged from 2.3–4.4, a range similar to, but somewhat narrower than, the range of 1.1–4.5 occupied by species in the family Labridae (Wainwright et al., 2002). Aspect ratio shows a strong correlation with swimming performance, with higher aspect ratio fins enabling higher critical swimming speeds than paddle-shaped fins (Walker and Westneat, 2002). Walker and Westneat (2002) showed that labriform swimmers with more elongate, wing-like fins and with a steeper (more dorsoventral) stroke plane can achieve and maintain higher swimming speeds than can labriform swimmers with lower-aspect-ratio paddle-like fins and shallower stroke planes.

This relationship between morphology and swimming performance is reflected at a broader level among the diversity of labrids by a strong association between habitat preference and fin shape. Recent analyses of habitat use in labrid fishes (Bellwood and Wainwright, 2001; Fulton et al., 2001) indicate that species with low aspect ratio fins are more abundant in deep, calm habitats, whereas taxa with high aspect ratio fins dominate shallow water habitats characterized by high water movement and the water column above the reef. The majority of species in this study have flapping fins that are rigid and have relatively high aspect ratios. Field observations of natural locomotor velocities and habitat use of the species described here could test these ecomorphological associations in other reef fish families.

The distribution of fin area along the fin axis in labrid fishes is correlated with stroke plane angle and locomotor performance during labriform locomotion (Walker and Westneat, 2002). Fin areas of reef fishes studied here (Table 2) show trends in fin area distribution that may also reflect fin stroke and performance differences. Species with high AR fins and a more dorsoventral flapping stroke have smaller fin tip areas and greater areas at the base compared to species with broader fins. Tip areas range from just 5–6% of total fin area in *Stethojulis* and *Gomphosus* up to 13–15% in *Thorichthys* and *Parupeneus*. Conversely, the fin base areas of *Abudefduf*, *Amphistichus*, *Gomphosus*, *Scarus*, *Stethojulis*, and *Zanclus* are over 20% of total fin area,

whereas *Parupeneus* and *Thorichthys* have about 15% of fin area at the base. Thrust forces from either lift, drag, or the acceleration reaction are determined by these fin areas and their oscillatory motion through the water (Drucker and Lauder, 2000; Walker and Westneat, 2000).

Fin ray morphology also reflects locomotor strategy among reef fishes. The first two fin rays of the pectoral fin are always nonbranched (Table 1), and these are usually joined to form the leading edge. Leading edge stiffness appears to be variable: high aspect ratio, wing-like fins are more stiff than broader fins. A stiff leading edge will enhance the development of flow circulation around the fin undergoing a steep flapping stroke with rapid stroke reversal (Walker and Westneat, 2002), although this remains to be tested with mechanical testing of fin ray stiffness. Interestingly, a number of the non-wrasse species have a nonbranched trailing edge ray. In an extreme case, in *Thorichthys meeki*, the first three rays and last three rays are nonbranched. This morphology, associated with a broader, more flexible fin, may provide for two "leading edges," one for normal fin stroke and the other for maneuvering and reversing.

The variation in bony processes at the bases of the fin rays is striking and has clear functional implications for flapping and rowing fins. Flappers have processes that are pronounced toward the leading edge and taper off toward the trailing edge. Rowers have pronounced processes along all rays, with special connections for the ABS and ABP that allow for control over a central pivot. Mechanical (force) advantage of fin rays can be expected to vary on a continuum between rowers and flappers; rowers should have a higher mechanical advantage, while higher performance swimmers should have a lower mechanical advantage with a concomitant increase in fin velocity advantage.

Structural variation in intraradial ligaments may also be functionally important. It is unknown how much variability in motion the different intraradial ligament configurations provide in a living fish. However, in preserved specimens the angle of rotation about each radial shows considerable variability. In some cases, all radials are orientated in the same direction, yielding uniform rotation for all fin rays (Fig. 9E; e.g., *Stethojulis trilineata*). In other cases, the third or fourth radial is orientated at a different angle than the first two, and, as a result, the first two-thirds of fin rays rotate over the first three radials and the last one-third of fin rays produce reduced rotation at a slightly different angle (Fig. 9A, *Abudefduf saxatilis*). If tendons leading to the trailing edge fin rays are actuated, the direction of rotation is up and forward as opposed to down and forward over the other radials. This motion may be used for fin motions that are observable during maneuvering or rapid stops.

Biomechanical models of fin morphology are currently under development and will provide a means for testing hypotheses generated by the diversity described above. The mechanical design of the pectoral girdle, musculature, and fin rays will be integrated in a mechanical model based on functionally important fin dimensions. This model will be used to quantify differences in mechanical advantage, velocity advantage, angle of attack, and stroke plane angle based on alternate fin designs. Such models are a useful tool for predicting the mechanical results of changes in morphology and motor patterns in musculoskeletal systems (Westneat, 1996). Comparisons of kinematics, muscle actions, skeletal mechanisms, and fin shapes in the context of mechanical models will be crucial for mechanical and evolutionary analyses of pectoral propulsion.

CONCLUSIONS

Many of the largest and most abundant families of coral reef fishes use their pectoral fins as their primary mode of locomotion. Little attention has been paid to the diversity and functional morphology of pectoral fin locomotion in reef fishes. Our results reveal previously unsuspected diversity in the musculoskeletal morphology of the pectoral girdle; this diversity has clear functional implications. However, predictions regarding muscle function and locomotor consequences of pectoral diversity must be tested with laboratory tests of swimming performance and fin physiology. When morphological and physiological approaches are combined, this research will be central to an understanding of reef fish locomotion. Therefore, fishes with labriform propulsion are an attractive target for evolutionary investigations aimed at understanding the evolution of functional systems in reef fish taxa. Although a higher-level phylogeny of labroids is elusive, and a phylogeny of perciform fishes is essentially nonexistent, the species from 10 families described here probably have evolved from divergent perciform lineages. Thus, it seems likely that labriform locomotion has evolved independently several times among reef fishes. Future research using resolved phylogenetic hypotheses can test hypotheses of the origin, diversification, and independent evolution of labriform propulsion.

ACKNOWLEDGMENTS

We thank P. Wainwright for assistance in collecting several of the specimens described here and J. Walker for discussion of pectoral fin shape analysis. For their insight and comments on the manuscript, we thank J. Fessler, J. Cooper, S. Watt, and two anonymous reviewers.

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APPENDIX

Raw values of pectoral girdle musculature masses measured in grams from the left side of specimens unless otherwise indicated

ID	Family	Genus	Species	TL	BM	ABS	ABP	ARRV	ADS	ADP	ARRD
FMNH 61658	Pomacentridae	<i>Abudefduf</i>	<i>saxatilis</i>	120	32.1	0.0540	0.0757	0.0220	0.0650	0.0703	0.0122
FMNH 61660	Pomacentridae	<i>Abudefduf</i>	<i>saxatilis</i>	128	43.3	0.0528	0.0673	0.0225	0.0602	0.0607	0.0098
FMNH 48155	Pomacentridae	<i>Abudefduf</i>	<i>saxatilis</i>	150	56.2	0.0587	0.0720	0.0240	0.0600	0.0564	0.0131
FMNH 42553	Embiotociade	<i>Amphistichus</i>	<i>argenteus</i>	109	16.4	0.0204	0.0385	0.0124	0.0191	0.0462	0.0061
FMNH 42588	Embiotociade	<i>Amphistichus</i>	<i>argenteus</i>	111	13.9	0.0238	0.0446	0.0125	0.0220	0.0367	0.0076
FMNH 42553	Embiotociade	<i>Amphistichus</i>	<i>argenteus</i>	115	19.4	0.0298	0.0467	0.0164	0.0231	0.0401	0.0100
LACM-W67-149A	Embiotociade	<i>Amphistichus</i>	<i>argenteus</i>	188	96.4	0.1283	0.2055	0.0772	0.1589	0.1877	0.0458
FMNH 44099	Pomacanthidae	<i>Centropyge</i>	<i>flavissimus</i>	71	8.6	0.0116	0.0140	0.0034	0.0076	0.0101	0.0019
FMNH 44099	Pomacanthidae	<i>Centropyge</i>	<i>flavissimus</i>	76	11.1	0.0153	0.0171	0.0055	0.0138	0.0138	0.0017
FMNH 16287	Pomacanthidae	<i>Centropyge</i>	<i>flavissimus</i>	122	49.1	0.0638	0.0777	0.0204	0.0484	0.0462	0.0061
FMNH 70267	Chaetodontidae	<i>Chaetodon</i>	<i>miliaris</i>	137	41.2	0.0560	0.0405	0.0116	0.0359	0.0340	0.0067
FMNH 70267	Chaetodontidae	<i>Chaetodon</i>	<i>miliaris</i>	146	55.8	0.0666	0.0552	0.0187	0.0396	0.0462	0.0075
FMNH 111974	Labridae	<i>Cheilinus</i>	<i>fasciatus</i>	230	234.6	0.1052	0.1448	0.0420	0.1239	0.2020	0.0282
FMNH 111972	Labridae	<i>Cheilinus</i>	<i>fasciatus</i>	247	224.7	0.1673	0.2832	0.0723	0.2190	0.2675	0.0473
FMNH 111973	Labridae	<i>Cheilinus</i>	<i>fasciatus</i>	260	389.2	0.2009	0.2902	0.0972	0.2805	0.3942	0.0610
FMNH 75894	Acanthuridae	<i>Ctenochaetus</i>	<i>striatus</i>	166	69.9	0.1686	0.2326	0.0762	0.1842	0.1753	0.0460
FMNH 75894	Acanthuridae	<i>Ctenochaetus</i>	<i>striatus</i>	178	77.9	0.2025	0.2378	0.0750	0.2142	0.1684	0.0473
FMNH 74045	Acanthuridae	<i>Ctenochaetus</i>	<i>striatus</i>	190	93.0	0.1839	0.2607	0.0906	0.2172	0.1769	0.0461
FMNH 111975	Labridae	<i>Gomphosus</i>	<i>varius</i>	198	76.0	0.0987	0.1293	0.0528	0.1096	0.2007	0.0605
FMNH 111977	Labridae	<i>Gomphosus</i>	<i>varius</i>	240	123.6	0.2427	0.4103	0.1523	0.2864	0.6172	0.1625
FMNH 111976	Labridae	<i>Gomphosus</i>	<i>varius</i>	243	126.0	0.1777	0.3150	0.1328	0.2659	0.5500	0.1660
FMNH 63635 L	Mullidae	<i>Parupeneus</i>	<i>pleurostigma</i>	174	52.9	0.0654	0.0722	0.0206	0.0718	0.1066	0.0155
FMNH 63635 R	Mullidae	<i>Parupeneus</i>	<i>pleurostigma</i>	174	52.9	0.0672	0.0753	0.0176	0.0733	0.1003	0.0116
FNQ-93-63	Labridae	<i>Scarus</i>	<i>frenatus</i>	213	165.2	0.1657	0.3358	0.1384	0.2589	0.3434	0.1392
FMNH 111980	Labridae	<i>Scarus</i>	<i>frenatus</i>	340	858.3	0.7188	1.3811	0.5118	1.0746	1.6951	0.4768
FMNH 110756	Labridae	<i>Stethojulis</i>	<i>trilineata</i>	120	22.5	0.0608	0.2487	0.0819	0.1092	0.2382	0.0760
FMNH 111978	Labridae	<i>Stethojulis</i>	<i>trilineata</i>	139	43.0	0.1006	0.4095	0.1363	0.1617	0.4251	0.1247
FMNH 111979	Labridae	<i>Stethojulis</i>	<i>trilineata</i>	148	48.9	0.1077	0.5499	0.1824	0.1886	0.5132	0.1640
FMNH 71263	Cichlidae	<i>Thorichthys</i>	<i>meeki</i>	125	22.1	0.0248	0.0112	0.0061	0.0187	0.0268	0.0022
FMNH 104554	Cichlidae	<i>Thorichthys</i>	<i>meeki</i>	130	32.2	0.0217	0.0128	0.0045	0.0153	0.0238	0.0017
FMNH 82601	Cichlidae	<i>Thorichthys</i>	<i>meeki</i>	133	36.4	0.0328	0.0195	0.0073	0.0237	0.0431	0.0030
FMNH 46078	Zanclidae	<i>Zanclus</i>	<i>cornutus</i>	124	27.3	0.0381	0.1240	0.0425	0.0491	0.1014	0.0266
FMNH 44225	Zanclidae	<i>Zanclus</i>	<i>cornutus</i>	137	61.0	0.4334	1.0876	0.3828	0.3455	0.9645	0.1683
FMNH 46078	Zanclidae	<i>Zanclus</i>	<i>cornutus</i>	155	90.6	0.5627	1.4683	0.5734	0.4276	1.5857	0.2922

TL, total length; BM, body mass; ABS, abductor superficialis; ABP, abductor profundus; ARRV, arrector profundus; ADS, adductor superficialis; ADP, adductor profundus; ARRD, arrector dorsalis. L, left pectoral girdle; R, right pectoral girdle.